And hence we have

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(f_{1,[\![\mathrm{AOB}]\!]}\rho_\beta\right) = \mathcal{L}_{\mathrm{LD}}^{\dagger}\left(f_{1,[\![\mathrm{ABO}]\!]}\rho_\beta\right) - \gamma\beta\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(U(q)\rho_\beta\right),$$

giving the general solution

$$f_{1,[\![\mathrm{AOB}]\!]} = -\frac{\beta}{2}\left(\gamma U(q) + \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right),$$

for the same choice of c.

Hence using the symmetry in the commutators we are able to solve for the leading order corrections for each of the schemes of the form $[\![\mathrm{XYZ}]\!]$. In summary, we find that

$$f_{1,[\![\mathrm{ABO}]\!]}(\boldsymbol{q},\boldsymbol{p}) = f_{1,[\![\mathrm{OAB}]\!]}(\boldsymbol{q},\boldsymbol{p}) = \frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) - \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25a}$$

$$f_{1,[\![\mathrm{OBA}]\!]}(\boldsymbol{q},\boldsymbol{p}) = f_{1,[\![\mathrm{BAO}]\!]}(\boldsymbol{q},\boldsymbol{p}) = -\frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) - \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25b}$$

$$f_{1,[\![\mathrm{AOB}]\!]}(\boldsymbol{q},\boldsymbol{p}) = -\frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) + \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25c}$$

$$f_{1,[\![\mathrm{BOA}]\!]}(\boldsymbol{q},\boldsymbol{p}) = \frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) + \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right), \tag{7.25d}$$

where the constant c is chosen to ensure each correction function has average 0, giving $c = \gamma\langle U(\boldsymbol{q})\rangle$. If required, we may continue this procedure in order to find correction functions for higher orders, though the added complexity means that we cannot always solve to find an explicit solution.

The results of our analysis suggest that to leading order the error scales linearly with the reciprocal temperature β, but more importantly a non-constant term appears which is of order γ. As our correction is of the form $\langle\vartheta f_1\rangle$ we can see that if $\vartheta = \vartheta_1(\boldsymbol{p})$ then this $\mathcal{O}(\gamma)$ term will not trouble us, as

$$\begin{aligned}\langle\vartheta_1(\boldsymbol{p})f_1(\boldsymbol{q},\boldsymbol{p})\rangle &= \pm\frac{\beta}{2}\left\langle(\gamma U(\boldsymbol{q}) - c)\vartheta_1(\boldsymbol{p}) \pm \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\vartheta_1(\boldsymbol{p})\right\rangle \\ &= \pm\frac{\beta}{2}\left\langle\boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\vartheta_1(\boldsymbol{p})\right\rangle,\end{aligned}$$

where the sign depends on the particular scheme used (which governs the form of f_1). Thus in the case of a function of pure momentum the order γ term does not

appear. However, if we instead set $\vartheta = \vartheta_2(\boldsymbol{q})$, then

$$\begin{aligned}\langle \vartheta_2(\boldsymbol{q}) f_1(\boldsymbol{q},\boldsymbol{p})\rangle &= \pm\frac{\beta}{2}\left\langle (\gamma U(\boldsymbol{q}) - c)\vartheta_2(\boldsymbol{q}) \pm \boldsymbol{p}^T \boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\vartheta_2(\boldsymbol{q})\right\rangle \\ &= \pm\frac{\gamma\beta}{2}\left\langle (U(\boldsymbol{q}) - \langle U(\boldsymbol{q})\rangle)\vartheta_2(\boldsymbol{q})\right\rangle,\end{aligned}$$

in which case our correction becomes $\mathcal{O}(\gamma)$. This is potentially disastrous if we choose the friction to be large (consider $\gamma \sim h^{-1}$), as the error introduced through discretization will dominate the true average to be sampled. Examining the algorithms themselves this is unsurprising, as although we have integrated the Ornstein-Uhlenbeck process exactly, when considering the limit of infinite friction in the ⟦XYZ⟧ schemes we obtain an inconsistent discretization. Hence these first-order schemes will be unsuitable for the integration of overdamped Langevin dynamics.

This is not always the case for first order schemes, however. For example the scheme denoted ⟦AOAB⟧ is consistent with the dynamics, and may be used for integrating the overdamped ($\gamma = \infty$) Langevin equation.

7.8.2 Numerical Experiment

Let us consider testing the computed first order corrections on a one dimensional example. Let $M = \beta = 1$ and define the biased double-well potential as

$$U(q) = (q^2 - 1)^2 + q/2.$$

We shall consider computing the average of a specific observable $\vartheta(q,p)$, and estimating the error due to discretization using the correction functions given in (7.25), where we choose the observable

$$\vartheta(q,p) = p^2 - qU'(q) + 2qp, \qquad \langle\vartheta\rangle = 0.$$

The low dimensionality of the problem and the simplicity of the observable and potential energy function make resolving the perfect sampling bias to leading order an easy task, allowing us to give a good estimate (to order h^2) of the discretization error. In Fig. 7.4 we plot the results of $\langle\vartheta\rangle_h$ computed via numerical experiment, using each of the six first order schemes at either $\gamma = 1$ or $\gamma = 2$, and compare them to our first order prediction

$$\langle\vartheta\rangle_h \approx \langle\vartheta\rangle + h\langle f_1\vartheta\rangle.$$

The numerical experiment at each timestep is averaged from 16 independent experiments, each running over a time interval of 10^8. The results match our